Joel Albert Montuya

Applied AI Engineer | Backend and Integration

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Eligible to work in Ireland (Stamp 1G)

Profile

Applied AI engineer with 2.5 years at Accenture building the backend and API integrations behind enterprise conversational-AI applications, including 13 live production use cases. I then completed an MSc in Data Analytics (2:1) and built and deployed RAG and agent systems on Azure and Kubernetes. I like connecting AI services to real business systems and checking that they work before release. Looking to join an AI R&D team doing exactly that.

Technical Skills

Languages	Python, JavaScript (Node.js), SQL, Kusto Query Language (KQL)
AI & data	RAG, agentic AI, LLM integration, LangChain, LangGraph, Microsoft Foundry, Azure AI Search, TensorFlow, Pandas, NumPy, Spark, Hadoop
Backend & integration	FastAPI, Django, Node.js, REST APIs, OpenAPI tool-calling, API integration, integration testing, PostgreSQL, MongoDB
Cloud & DevOps	Azure, Docker, Kubernetes (GKE), GitHub Actions CI/CD, Git & GitHub

Projects

Uhie: Home Care Operations Agent

Jul 2026 – Aug 2026

Microsoft Foundry · Azure AI Search · React · Node.js · MongoDB · Vercel · Supabase
GitHub · Live Demo

- Built a Microsoft Foundry agent inside a live healthcare web app, as the capstone for Microsoft's AI-103 course
- Added RAG over HR and care-policy documents with Azure AI Search, so answers are grounded and cited
- Connected the agent to the platform with OpenAPI tool calls (filing leave, finding a suitable carer, checking schedules), with admin and caregiver permissions enforced on the backend

Flexible RAG Platform

Aug 2026

Python · FastAPI · ChromaDB · Ollama · Docker · Kubernetes · GKE · GitHub Actions
GitHub

- Built a data-agnostic RAG platform on FastAPI, ChromaDB and Ollama, with runtime PDF ingestion, vector indexing, authenticated retrieval and cited answers
- Deployed it to GKE as separate containers for the API, ChromaDB and Ollama, with persistent volumes and GitHub Actions CI/CD

MSc Capstone: Fine-Grained Gym Exercise Recognition

Feb 2025 – Jun 2026

TensorFlow · Deep learning · Signal preprocessing and augmentation

- Compared five architectures (CNN, LSTM, GRU, CNN-BiLSTM, CNN-ResBiGRU) on accelerometer and gyroscope data
- Reached 97.05% test accuracy with CNN-BiLSTM and a 96.72% F1-score with CNN-ResBiGRU; CNN was fastest at 246.91 ms inference

Experience

AI Engineering Analyst, Accenture, Philippines

Jun 2022 – Aug 2023

- Built Node.js backend features for enterprise conversational-AI applications, turning business and conversation requirements into deployable components
- Integrated chatbot services with enterprise APIs and supported integration testing, troubleshooting, release readiness and production deployment
- Built security-verification flows for telecoms use cases, including SIM-swap checks and live-agent handover
- Used Azure KQL on production queries to track daily active users and find failing API endpoints, then built dashboards that reduced downtime and sped up issue resolution across 13 live use cases
- Built custom paginated service-list workflows beyond the framework's standard features, supporting end-to-end customer purchase journeys
- Reviewed code and ran knowledge-transfer sessions for associate engineers

AI Engineering Associate, Accenture, Philippines — Feb 2021 – Jun 2022

- Turned conversational designs into backend implementations and supported API integration, testing, deployment and troubleshooting on enterprise chatbots
- Ranked first overall in Accenture bench training (RPA, machine learning, chatbots, computer vision) and passed the Java Spring Boot backend assessment
- Won the FY22 Q1 Gantimpala Award, Agila (Eagle), and was promoted to Analyst in about a year

Healthcare Assistant and Shift Team Leader (part-time), Belmont, Ireland — Nov 2023 – Jun 2026

- Worked part-time alongside full-time MSc studies
- Led shift teams, coordinating schedules and delegating tasks to keep service on track under time pressure

Education

MSc Data Analytics (NFQ Level 9, 2:1), CCT College Dublin — Feb 2025 – Jun 2026

Diploma in Project Management (Distinction), CCT College Dublin — Aug 2026

Diploma in DevOps (Distinction), CCT College Dublin — Aug 2025

BSc Computer Engineering, University of St. La Salle, Philippines — 2012 – 2019

Certifications

Microsoft Certified: Azure AI Apps and Agents Developer Associate — Sep 2026

- Credential ID 8CA4FX-2731F5, valid until September 2027 · Verify credential

Professional Development

- **AI-103: Developing AI Apps and Agents on Azure**, Udemy (Jul 2026 – Aug 2026): Microsoft Foundry, agentic solutions, RAG, MCP servers, Azure Cosmos DB, Azure AI Search, Azure Functions, Azure Container Apps, Blob Storage
- **Deloitte Australia Data Analytics Job Simulation**, Forage (Aug 2026): built a Tableau dashboard and used Excel to classify data and draw business insights
- Also: Google Introduction to AI Agents; Designing Agentic Systems with LangChain; RPA; Computer Vision (OpenCV); Machine Learning

Memberships

Machine Learning Dublin · Data, Cloud and AI in Dublin · Dublin AI Developer Group